T2D Glucose Digital Twin — System Architecture

End-to-End Pipeline: Observable Ingestion → UKF S3 Prior → PR-TFT Residuals → SSE Proxy → Workstation

TIER 1: MULTIMODAL OBSERVABLE INGESTION (CONTRACT S3)

Static EHR Context
Age, BMI, HbA1c, eGFR, diabetes duration, medication flags (metformin, SGLT2i, GLP-1, insulin).

Dynamic CGM Telemetry
5-minute continuous glucose readings (mg/dL), trend velocity (Δmg/dL/min), sensor flag.

Cardiorespiratory Streams
Heart rate (HR), Heart rate variability (HRV), resting HR warm-up calibration.

Behavioral & Dietary Logs
Logged meal carbs (grams), meal timing, step counts, sleep duration/stages.

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TIER 2: HYBRID PHYSIOLOGICAL & DEEP LEARNING ENGINE

Mechanistic UKF S3
2-compartment Bergman minimal model + gut absorption dynamics. Produces 120-min physiological prior.

PR-TFT Encoder/Decoder
Variable selection networks, LSTM temporal encoder, causal multi-head self-attention.

Residual Quantile Heads
Learns residual deltas over UKF prior. Outputs 7 quantiles (p05, p10, p25, p50, p75, p90, p95).

Spike Hazard Predictor
Predicts discrete-time spike onset hazard over 120-minute horizon. Calibrated isotonic risk.

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TIER 3: RELIABLE SERVING & TELEMETRY PROXY

FastAPI Inference Service
serve_pr_tft.py (port 8001). Generates ForecastResponse v1 and stateful alerts.

Next.js SSE Proxy Route
createSafeSSETransformStream (port 3000). Chunk-invariant earliest-delimiter parser.

Client Telemetry Hook
useRealtimeTelemetry. Bounded exponential backoff, envelope validation, contract enforcement.

What-If Simulation API
Counterfactual meal/activity simulation via /api/forecast. UI editor explicitly disabled.

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TIER 4: CLINICAL WORKSTATION DASHBOARD

Cohort Directory

Patient Workspace

Risk Waterfall

Model Contributions

/cohort page. Search, filter by status (Healthy, Prediabetes, T2D), select subjects.

/patients/[id]. Visualizes glucose ribbon, current trend, and alert status.

Cumulative adverse event probability curve crossing clinical thresholds.

Arithmetic decomposition: UKF baseline, habitual meals, logged carbs, deep residuals.